



FIGURE 12: The object is not in fact a lens cap. These two photos illustrate scale invariance: the terrain is fractal. Compare it to man-made objects such as a car or a house. *Source: Professor Stephen W. Wheatcraft, University of Nevada, Reno.*

In the rest of this chapter I will explain how I can endorse Mandelbrotian fractals as a representation of much of randomness without necessarily accepting their precise use. Fractals should be the default, the approximation, the framework. They do not solve the Black Swan problem and do not turn all Black Swans into predictable events, but they significantly mitigate the Black Swan problem by making such large events conceivable. (It makes them gray. Why gray? Because only the Gaussian give you certainties. More on that, later.)

THE LOGIC OF FRACTAL RANDOMNESS (WITH A WARNING)*

I have shown in the wealth lists in [Chapter 15](#) the logic of a fractal distribution: if wealth doubles from 1 million to 2 million, the incidence of people with at least that much money is cut in four, which is an exponent of two. If the exponent were one, then the incidence of that wealth or more would be cut in two. The exponent is called the “power” (which is why some people use the term *power law*). Let us call the number of occurrences higher than a certain level an “exceedance”—an exceedance of two million is the number of persons with wealth more than two million. One main property of these fractals (or another way to express their main property, scalability) is that the ratio of two exceedances* is going to be the ratio of the two numbers to the negative power of the power exponent. Let us illustrate this. Say that you “think” that only 96 books a year will sell more than 250,000 copies (which is what happened last year), and that you “think” that the exponent is